

BetterPresser — User Guide & Operations Manual

Product: BetterPresser (VST3 & Standalone)

Developer: Archie DSP

Engine: 64-bit Floating-Point DSP

Assignment Deliverable: Operations Manual

1. Overview & Introduction

BetterPresser is a high-precision, 64-bit transparent digital audio compressor designed for mixing, mastering, and live sound production. Combining an intuitive vintage hardware console aesthetic inspired by classic analog processors with mathematical digital transparency, BetterPresser delivers clean dynamic control without unwanted harmonic distortion or coloration.

Key Features

- **Transparent Dynamic Control:** Clean feed-forward gain computer with continuous second-order polynomial soft-knee.
- **Dual Detection Engines:** Instantaneous **Peak** detection for aggressive transient control vs. continuous **RMS** energy integration for smooth vocal and bus leveling.
- **Dual-Mode Visualizer Display:**
 - **Analog VU Meter:** Illuminated vintage dial with physical 2nd-order mass-spring-damper needle ballistics.
 - **Live Scrolling Graph:** Smooth, eye-comfortable waveform visualizer displaying Input, Output, and Gain Reduction curves over a ~3.3-second history window.
- **Precision Sidechain High-Pass Filter:** 2nd-order SVF (20 Hz – 500 Hz) to eliminate sub-bass pumping, with a dedicated **Sidechain Listen** monitor mode.
- **Parallel Compression:** Sample-exact Dry/Wet crossfader for New York style parallel compression.
- **Precision Level Metering:** Vertical stereo Input and Output level meters with peak hold bars and clip warning indicators.

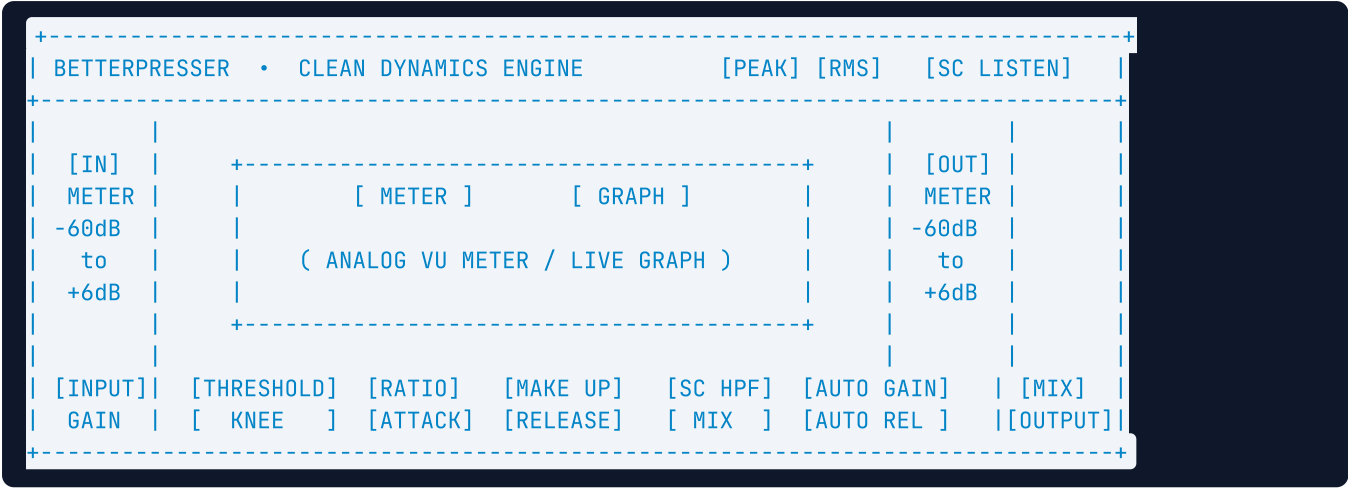
2. Installation & DAW Compatibility

Windows VST3 Path: `C:\Program Files\Common Files\VST3\BetterPresser.vst3`

Standalone Application: `build/BetterPresser_artefacts/Release/Standalone/BetterPresser.exe`

Compatible with all major 64-bit VST3 digital audio workstations including **Ableton Live**, **Cockos Reaper**, **FL Studio**, **Logic Pro**, **Steinberg Cubase**, **Studio One**, and **Bitwig Studio**.

3. Control Interface & Layout



4. Parameter Reference Guide

Control	Parameter ID	Range	Default	Description
Detection Mode	<code>detectionMode</code>	Peak / RMS	Peak	Selects instantaneous peak or energy-averaged RMS detection.
Threshold	<code>threshold</code>	-60.0 dB to 0.0 dB	-20.0 dB	Sets the decibel level at which gain reduction begins.
Ratio	<code>ratio</code>	1.0:1 to 30.0:1	4.0:1	Determines how much compression slope is applied above threshold.
Attack	<code>attack</code>	0.1 ms to 200.0 ms	15.0 ms	Controls how rapidly gain reduction engages when crossing threshold.
Release	<code>release</code>	5.0 ms to 2000.0 ms	100.0 ms	Controls recovery speed back to unity gain after signal drops below threshold.
Auto Release	<code>autoRelease</code>	Off / On	Off	Dynamically adapts release time (fast for transients, smooth for sustained tones).
Knee	<code>knee</code>	0.0 dB to 24.0 dB	6.0 dB	Width of the polynomial soft-knee transition zone for artifact-free compression.
SC HPF	<code>sidechainHPF</code>	20 Hz to 500 Hz	20 Hz (Off)	2nd-order high-pass filter on sidechain path to prevent bass pumping.
SC Listen	<code>sidechainListen</code>	Off / On	Off	Auditions the filtered sidechain detection signal through the outputs.
Make Up	<code>makeUpGain</code>	-24.0 dB to +24.0 dB	0.0 dB	Manual output gain compensation applied to the wet signal.
Auto Gain	<code>autoGain</code>	Off / On	Off	Automatically calculates and applies mathematical makeup gain.
Mix	<code>mix</code>	0.0% to 100.0%	100.0%	Crossfades between dry input (0%) and compressed audio (100%).
Input Gain	<code>inputGain</code>	-24.0 dB to +24.0 dB	0.0 dB	Pre-compression trim gain.
Output Gain	<code>outputGain</code>	-24.0 dB to +24.0 dB	0.0 dB	Master output volume trim.

5. Practical Mixing Recipes & Presets

1. Punchy Snare & Drum Bus

Detection: Peak • **Threshold:** -18 dB • **Ratio:** 4.0:1 • **Attack:** 25 ms • **Release:** 80 ms • **Knee:** 3 dB • **SC HPF:** 60 Hz • **Mix:** 100%

Result: Lets the initial drum punch hit hard while clamping down on the ringing tail for punch and presence.

2. Transparent Lead Vocal Leveling

Detection: RMS • **Threshold:** -24 dB • **Ratio:** 2.5:1 • **Attack:** 10 ms • **Release:** 150 ms • **Auto Release:** On • **Knee:** 12 dB • **Auto Gain:** On • **SC HPF:** 100 Hz

Result: Smooth, invisible dynamic riding that glues the vocal at the front of the mix without audible pumping.

3. Master Bus Glue

Detection: RMS • **Threshold:** -12 dB • **Ratio:** 1.5:1 to 2.0:1 • **Attack:** 30 ms • **Release:** 100 ms • **Knee:** 6 dB • **SC HPF:** 85 Hz • **Mix:** 100%

Target: Aim for 1.0 to 2.5 dB of Gain Reduction on the analog VU meter to knit the mix elements together.

4. Aggressive New York Parallel Compression

Detection: Peak • **Threshold:** -32 dB • **Ratio:** 12.0:1 • **Attack:** 0.5 ms • **Release:** 40 ms • **Knee:** 0.0 dB • **Mix:** 35%

Result: Heavily crushes drum transients in parallel, injecting massive room body underneath the dry punch.

6. Shortcuts & Interaction

- **Fine-Tune Parameter:** Hold **Shift** while dragging any rotary knob for 10x high-precision adjustment.
- **Reset to Default:** **Double-Click** any knob to immediately return it to its default setting.